

Chapter 137

Secondary Qualifiers

Secondary correction describes isolating a specific part of the image, or a specific subject, using a key.

Keys in DaVinci Resolve are grayscale images that define which areas of the picture you want to alter (in white) and which parts of the picture you want to leave alone (in black).

Keys can be generated either using the controls in the Qualifier palette, by using a Power Window, or by importing an external matte,

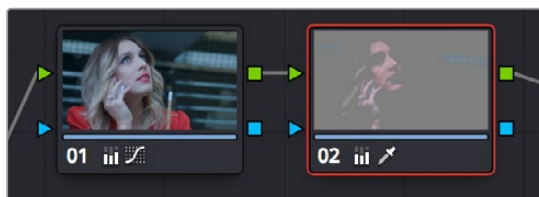
For more information on how to use external mattes, see *Chapter 146, “Combining Keys and Using Mattes.”* This chapter focuses on how you can use qualifiers to key a range of color values (similarly to doing a green screen key) in order to create a matte with which to do this kind of isolated adjustment.

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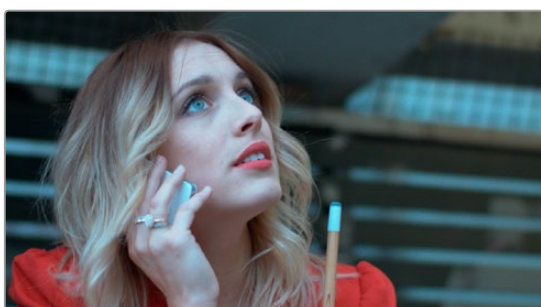
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Secondary Qualifiers

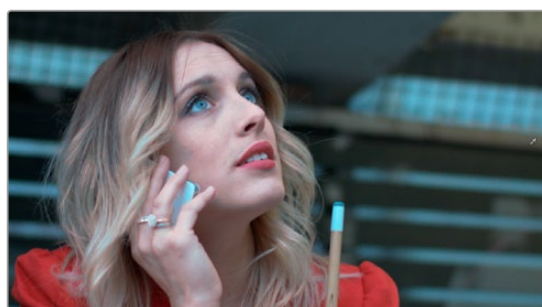
This section covers the use of the Qualifier palette, which lets you pull a 3D, HSL, RGB, or Luma key, with which to isolate the correction you need. The Qualifier controls are fast and flexible when you need to isolate an irregularly shaped subject with a distinct range of color or lightness. Since you're generating a key by sampling the image, there's no need for tracking or keyframing, so in the right situation qualifiers can be your fastest solution. In the following example, the client likes the slightly cool treatment overall, but wishes that the skin tones were a little more vibrant. This is exactly the sort of situation where qualifiers can help out. Adding a second node, and using the 3D or HSL Qualifier to isolate the face makes it relatively simple to add color exactly where you want it.



Adding a second node and using HSL Qualification to isolate the skin tones



The image with a simple primary correction



The final adjusted image

Just about every control in the Color page can be limited using of the Qualifier modes available in DaVinci Resolve. This makes the Qualifier palette a jack-of-all-trades tool with 101 uses. A few practical examples include keying a red element that's too intense for broadcast, in order to darken it or desaturate it; keying a range of green foliage, so you can shift its hue to a more attractive color; keying an actor's skin tone in a commercial, to apply some selective softening to it; or keying a range of sky, in which to add blue.

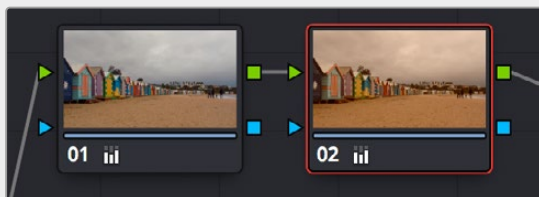
The Qualifier palette is color-space-aware when you're using Resolve Color Management (RCM) or ACES. This enables Qualifiers to create high-quality keys as you would expect, no matter what the color space of the original media is, or what Timeline Color Space you're using, for both SDR and HDR mastering. This makes Qualifier isolations easier, and a more consistent experience no matter what your workflow happens to be.

Adding a Secondary Operation to the Node Editor

Any node can be switched between functioning as a primary correction, where the adjustments you make affect the entire image, and a secondary correction, where you're adjusting a specific element in the scene. The only difference is that nodes being used for secondary corrections are limited using a qualifier, Power Windows, or an external matte.

If you're planning to add a secondary operation to the current grade, you'll need to first add another node in the Node Editor. When pulling a key to qualify part of the image, it's important to understand that you'll be sampling the YRGB values being fed to that node from any previous nodes in the tree. That means that the state of the image being fed to a node you're qualifying affects the key you're pulling.

For example, if the image coming out of Node 1 is well saturated and has a neutral color balance with a wide range of colors, but the image coming out of Node 2 applies a low saturation, monochromatically orange color wash, you may find it more difficult to pull a detailed key from Node 2 than you would from Node 1.



Choosing your battles—the image coming out of Node 1 will be easier to key than the highly stylized image coming out of Node 2

This is important because you have the flexibility of determining from what image you want to try pulling a qualified key. By connecting the node that's outputting the best YRGB image for the key you're trying to create to the node you're qualifying, you can control what you're keying.

For more information about choosing which node to use for setting up a qualifier, see *Chapter 146, "Combining Keys and Using Mattes."*

The Qualifier Interfaces

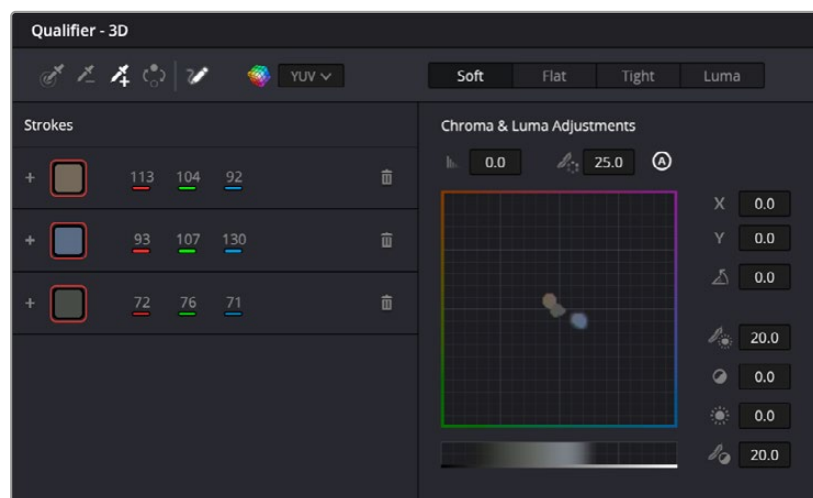
The DaVinci Resolve Qualifier palette interface is straightforward. To the left, graphical controls above numeric parameters let you manually adjust what ranges of each color component contributes to the key you're creating. To the right, Selection Range tools below let you define a key by sampling pixels of the image using the pointer, while below a set of Matte Finesse parameters let you alter the shape of the key that's been pulled.



Qualifier palette with HSL controls selected

The default qualification mode is the HSL Qualifier, which uses three color components, hue, saturation, and luma, to define a key. However, you can also use the RGB or LUM (Luma) qualification modes to pull keys using other combinations of color components. The LUM qualifier mode, in particular, lets you make targeted adjustments to specific ranges of image lightness. This is a technique employed by many colorists to alter color temperature within a specific range of image highlights or shadows.

Alternately, you can use the 3D qualifier to quickly and easily pull well-refined keys by drawing lines to sample colors from the image that correspond to volumes of color within a three-dimensional gamut. While the underlying technology is sophisticated, all you have to do is to draw blue lines to sample colors you want to isolate, or red lines to sample colors you want to subtract from the isolation you're creating, all of which automatically generate a high-quality key. Each line you draw adds a sample to the selection list; you can turn each sample off and on to evaluate its contribution to the resulting key, or delete samples that don't make a positive contribution.



The 3D qualifier

No matter which qualifier mode you use, the Matte Finesse controls make it easy to refine the resulting key to be even cleaner and more usable. In some instances, you can even take a marginal key that would otherwise be unusable, and squeeze it into something useful using the Clean Black, Clean White, and Blur Radius controls.

Which Qualifier Do I Use?

The Qualifier palette's four modes offer you the flexibility to use the best keyer for the job when it comes to isolating a range of color or brightness values. In some cases, keys that are difficult to pull using some modes are easier to pull using others. Here's a brief summary:



HSL: In many instances, the HSL keyer is not as immediately accurate as the 3D keyer, and will include a broader portion of the image for any given sample. On the other hand, if the 3D keyer is not giving you satisfactory results for a particular shot, the HSL keyer can sometimes do a better job. Because of its interface, the HSL keyer makes it easier to “fine-tune” the range and softness of each individual color component that's sampled, in order to improve the result. The HSL keyer also gives you the option of disabling color components that you don't want to contribute to the final key, so that you can pull a saturation-only key, or a hue-only key, for instances where that may solve the issue at hand.



RGB: The RGB keyer shares many of the limitations and advantages of the HSL keyer, but since you're sampling and adjusting red, green, and blue color components, the specificity with which you can fine-tune the resulting key works much differently.



LUM: The LUM keyer works specifically to isolate parts of the image based on image tonality, lightness, or darkness. This is the perfect tool when you're trying to isolate highlights or shadows in the image, which let you solve a multitude of different creative and technical issues. The LUM keyer is identical to using the HSL keyer with the Hue and Saturation qualifier controls turned off.



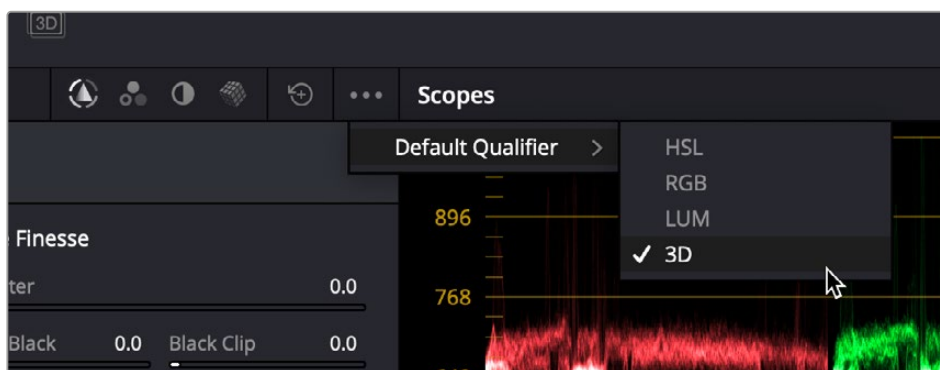
3D: The 3D keyer is a good one to start with if you're trying to isolate a range of color such as a blue shirt, a cyan sky, a performer's skin tone, or the orange leaves of a tree in autumn. Its interface of drawing lines over the part of the image you want to isolate, coupled with its high quality and extreme specificity, make it a fast and accurate tool to use in a variety of circumstances. However, the 3D keyer always samples every color component of the image; it's not useful when you want to isolate specific color components, such as luma only, or hue and saturation without luma. The 3D keyer's greatest strength, the speed with which you can sample areas of the picture to include (or exclude) from the final key, is also occasionally a weakness with images where your initial samples aren't giving you satisfactory results, because there aren't many ways to fine tune the key as it's being generated (although you can manipulate the result). On the other hand, for most images you would want to qualify, two or three samples is all you need, in conjunction with using the Matte Finesse controls to adjust the resulting key. If you need to do some compositing in the Color page, the 3D keyer also does an excellent job doing blue and green screen keying to create transparency, and has a built-in Despill control as well.

The following two sections provide an overview of how to use the 3D and HSL keyer modes in DaVinci Resolve.

Set the Default Qualifier Mode

You can set your favorite qualifier to open as the default when you apply a qualifier on a new clip on the timeline. This can save a significant amount of clicks over the course of a project. As expected, if a qualifier other than the default has been applied to the clip, that qualifier will open instead.

To change the Default Qualifier, click on the Qualifiers (3-dot) Option menu, select Default Qualifier, and choose an option from the list.



Changing the Default Qualifier to 3D, which will now open the 3D qualifier instead of the HSL qualifier on new clips

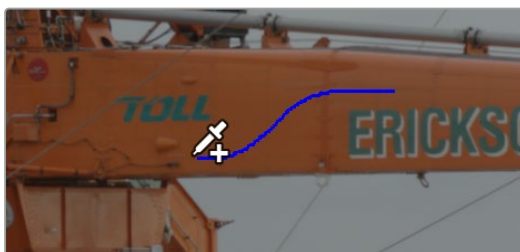
Basic Qualification Using the 3D Qualifier

The 3D Qualifier mode offers a fast, simple way of pulling a key to isolate a range of color in the image, by drawing lines over the parts of the image you want to key. Each line you draw over the image adds to or subtracts from the cloud of values you're carving out of a three-dimensional representation of all available colors; you don't see this representation, but this "under the hood" functionality is what gives the 3D keyer its name.

The 3D keyer is a general-purpose keyer, letting you isolate any color you like. However, the 3D keyer is not good at pulling luma-only keys. If you're looking to isolate a range of luma values in the image, you should use the LUM mode.

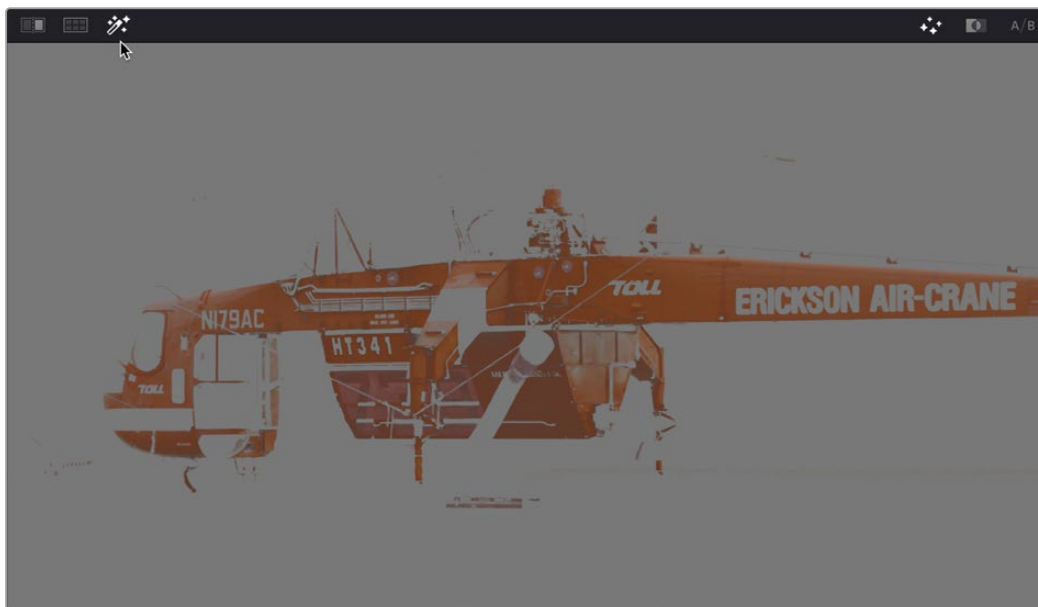
To use the 3D mode to isolate a subject in the Viewer:

- 1 Open the Qualifier palette, choose the 3D icon in the upper right and click the eyedropper.
- 2 Click and drag across the part of the image you want to isolate to draw a line. Lines that add to the key are colored blue.



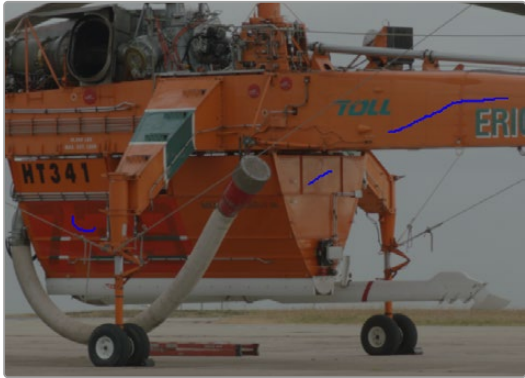
Drawing a line to create a key using the 3D qualifier

- 3 To see the key you're creating as you work, click the Highlight button in the Viewer Options toolbar at the top of the Viewer. As you draw the line, the viewer will update to show your current key in real time.



Viewing the key using the Highlight button

- 4 If necessary, draw additional lines to add more of the image to the key you're creating. Ordinarily, it's a good idea to not to draw more than two or three lines to sample the part of the image you're trying to isolate. A few long strokes are better than several short ones. Ideally, you want the key you're pulling to have a somewhat soft edge, which makes it easier to use the Matte Finesse controls to fine-tune your result later on. If you draw too many lines to sample the image, the result can be a key with hard, jagged edges that can sometimes be more difficult to adjust later on.



A hard-edged key created with three samples

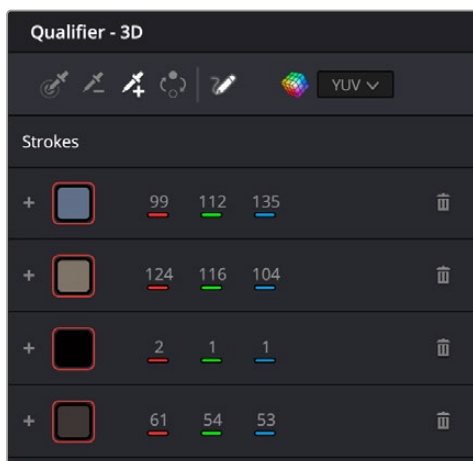


- 5 Alternately, if there are parts of the image that are included in the key that you want to omit, hold the Option key down and draw a line over these areas. You can see that Option-drawn lines are red, and these subtract areas from the key you're creating. Again, try not to overdo drawing subtractive red lines, as you can end up with an overly hard-edged key that's difficult to adjust using other controls in the next few steps. Additionally, you can still use power windows in conjunction with the 3D qualifier to exclude specific parts of the image.



Blue lines add to the key and a red line subtracts the grass from the key, as seen with the highlight turned off

- 6 If you find yourself wondering whether a particular sample in the list at left is doing any good, you can click the color patch at the left of any entry in the list to toggle that sample off and on. If you decide you don't need a particular sample, click the trash can button at the right to delete it.



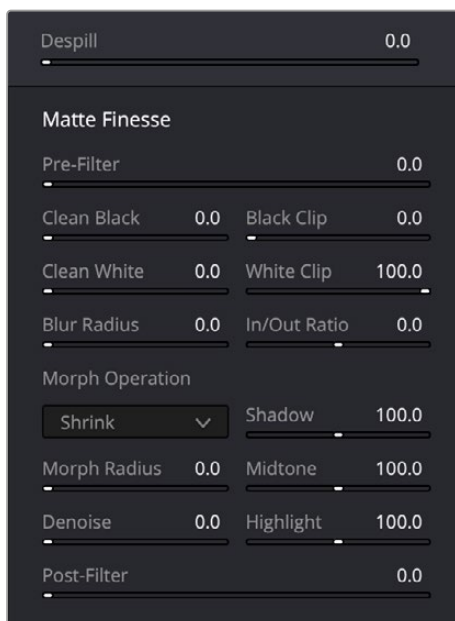
Controls for toggling samples off and on, and deleting them from the list

- 7 When you're satisfied that the key is good enough to start fine-tuning, stop drawing lines. It's all right if there's a bit of noise, speckling, or if there are small holes in the key you're creating, because you can take care of these using the Matte Finesse controls.



A sampled key that's ready for further refinement

Nearly every key you pull will benefit from some further "post-key" refinement. What this means is that, once you've created the best key you can procedurally through sampling the image, you can now adjust the resulting key itself, which is just a gray-scale image, to improve the isolated result. This is what the Matte Finesse controls are for.



Matte Finesse controls

- 8 The three most common methods of key refinement using the Matte Finesse controls are to increase Clean Black to fill in “holes” in the parts of the image you’re omitting (the background), increase Clean White to close holes in the part of the image you’re isolating (the foreground), and then use Blur Radius and In/Out Ratio controls to blur the edge of the key and push it in and out. Using these controls, you can vastly improve nearly any key without the need to endlessly readjust the Qualifier controls.

More information about using the Matte Finesse controls appears later in this chapter.

- 9 When you’re finished, click the Highlight control to turn the highlight off, and make whatever adjustment you need. In this example, the orange of the helicopter will be adjusted using the Hue control to make it blue instead.



Adjusting the orange color of the helicopter to be blue instead

Basic Qualification Using the HSL Qualifier

The HSL qualifier is one of the most flexible tools in DaVinci Resolve. Using any combination of sampled hue, saturation, and luma, you can isolate different ranges of color and contrast for specific adjustment. HSL is a general-purpose keyer, meaning you can key any color you like; it's not restricted to blue or green.

The most straightforward way to use any qualifier is to sample the image in the Viewer. A set of four controls let you sample pixels within the image in different ways. When you sample one or more pixels, the Hue, Saturation, and Luma values are analyzed, and set different ranges in each of the Hue, Saturation, and Luma qualifier controls. There are a number of different eyedropper controls available to control how you sample color in the image.

- **Sample Eyedropper:** The first control you use whenever sampling the image. It defines the initial range of the qualification; you must use this tool before any of the others. Clicking once with this tool selects a single pixel value, while clicking and dragging selects a range of image values that add together.



Color
Picker button

- **Add/Subtract Color Range:** These two controls let you add areas of the image to, or subtract them from, the currently selected inner range of values that define the core of the key. As with the eyedropper, you can click on single pixels, or drag over a range of color.



Add and Subtract
color buttons

- **Add/Subtract Softness:** These two controls let you redefine the softness that transitions from the inner range of the key, falling off towards the outer edge of the key. Just like the other controls, you can click on single pixels, or drag over a range of color.



Add and Subtract
qualifier softness

NOTE: When you select one of the sampling controls, it remains selected until you select another sampling control. If you select another clip in the Timeline, then click in the Viewer to sample a key and nothing happens, make sure that you select the Eyedropper control in the Qualifier palette.

If you're working with the DaVinci Resolve onscreen interface, the easiest way to qualify a subject is to use the pointer in conjunction with the sampling controls.



Example image to qualify

To use the HSL Qualifier mode to isolate a subject in the Viewer:

- 1 Open the Qualifier palette, choose HSL from the mode drop-down, and click the eyedropper.
- 2 Either click a pixel of the part of the image you want to isolate, or click and drag across a range of pixels within that subject.



Picking to qualify the ocean

- 3 To see the key you're creating as you work, click the Highlight button in the Viewer Options toolbar at the top of the Viewer.

By default, the area of the image you're isolating appears in color, while the area of the image being excluded appears solid gray.

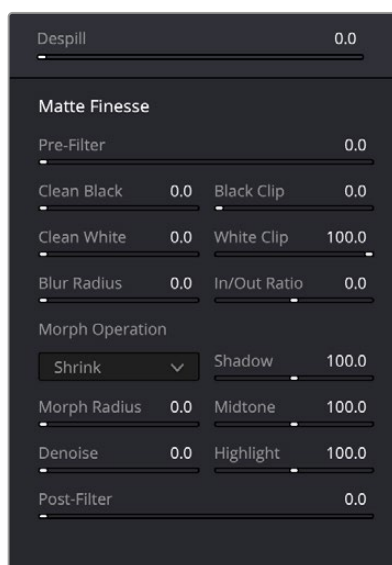


Viewer image with highlight turned on, isolated region appears in color, omitted region appears gray

Even though the highlight obscures the rest of the image, this doesn't prevent you from using the sampling tools to select pixels you can't see.

- 4 To subtract from the inner range of the key you're creating, click the minus Color Range control, and click or drag across the portion of the keyed image that you'd like to exclude.
- 5 To add softness to the outer range of the key you're creating, click the plus Softness control, and then click or drag across the portion of the image you'd like to include as a soft edge.

Once you've finished using the range and softness sampling controls to refine the key you're using to isolate the subject, nearly every key you pull will benefit from some further "post-key" refinement. What this means is that, once you've created the best key you can procedurally through sampling the image, you can now adjust the resulting key itself, which is just a grayscale image, to improve the isolated result. This is what the Matte Finesse controls are for.



Matte Finesse controls

- 6 The three most common methods of key refinement using the Matte Finesse controls are to increase Clean Black to fill in “holes” in the parts of the image you’re omitting (the background), increase Clean White to close holes in the part of the image you’re isolating (the foreground), and then use Blur Radius and In/Out Ratio controls to blur the edge of the key and push it in and out. Using these controls, you can vastly improve nearly any key without the need to endlessly readjust the Qualifier controls.

More information about using the Matte Finesse controls appears later in this chapter.

- 7 When you’re finished, click the Highlight control to turn the highlight off, and make whatever adjustment you need. In this example, the saturation has been raised and the color balance changed to make the adjustment obvious.



Altered color balance of the ocean and sky

HSL Qualifier Presets

There is a set of HSL Qualifier presets, or Six Vector presets, that you can use to isolate automatically a predefined wedge of color (green, blue, red, magenta, yellow, or cyan) or chroma intensity (chroma light, chroma dark, or both). These presets are available either from the Color > Presets menu, or using the DaVinci Resolve Advanced control panel.

Selecting one of these presets automatically sets the HSL Qualifier to the corresponding range or color or lightness. Afterwards you can continue modifying the HSL Qualifier to suit your purpose.

Six Vector - Green
Six Vector - Yellow
Six Vector - Red
Six Vector - Magenta
Six Vector - Blue
Six Vector - Cyan
Chroma Light
Chroma Dark
Chroma Light and Dark

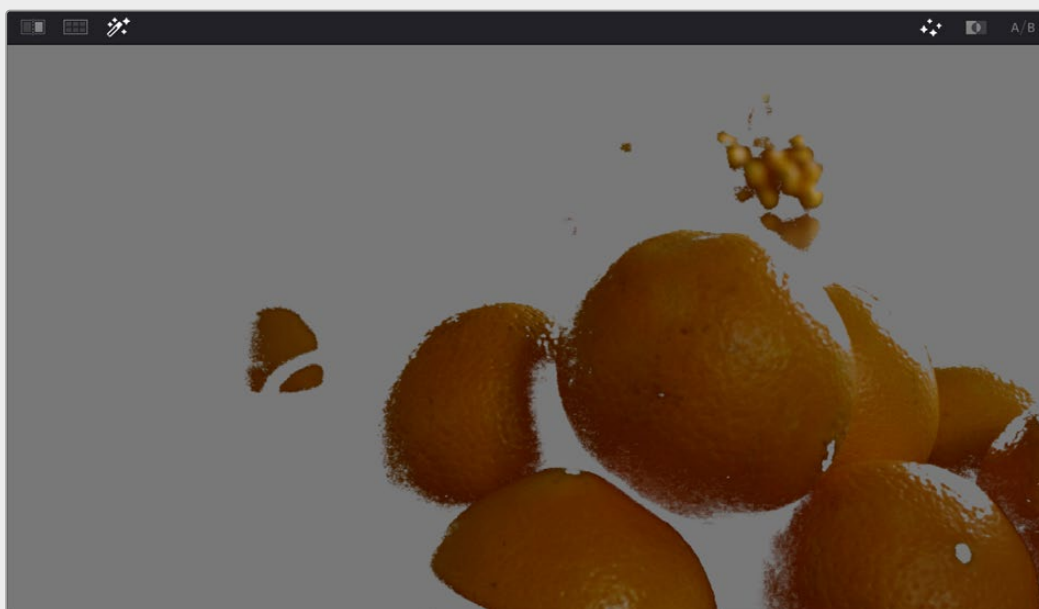
The Color > Presets menu

Using Highlight to See What You're Isolating

The Highlight control at the top of the Viewer (revealed by turning on Viewer Options in the Viewer's Option menu) lets you overlay a representation of the key you're creating over the current image in the Viewer. Overlays are also mirrored to your video output, so they'll show up on an external display as well, in the event that you're hiding the onscreen Viewer.

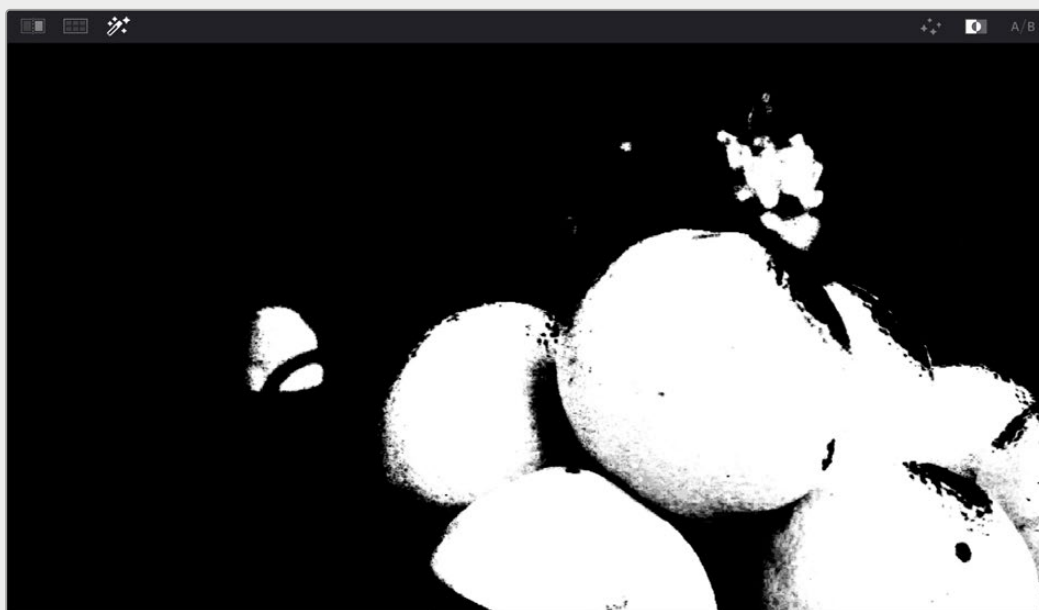
There are two types of highlights with which you can evaluate a qualification you're creating. Each is useful for different tasks.

Flat-Gray: The default highlight that DaVinci Resolve uses shows the selected portion of the image with the original colors, and the unselected portion of the image with a flat gray. It is useful for seeing the subject you're isolating even while you're adjusting the Qualifier controls, which lets you also see the color or contrast adjustments you're applying at the same time.



Oranges qualified and highlighted

High-Contrast: Pressing Option-Shift-H, you can show what's called a "high-contrast" highlight, which should be familiar to you if you've used other color correction applications and plugins; it's a more typical display wherein the selected portion of the image is white, and the unselected portion of the image is black. The high-contrast highlight is useful in situations where you need to eliminate holes in a key, or evaluate how "chattery" a key is since irregularities are easier to spot when divorced from the original image.



High-Contrast B&W highlighting

To enable or disable a highlight, do one of the following:

- In the Viewer Options toolbar at the top of the Viewer, click the Highlight control, and choose the type of highlight you want from the buttons that appear at the right.
- Choose an option from the Highlight submenu in the Viewer's option menu; a regular Highlight is Shift-H.

To enable or disable a high-contrast highlight, do one of the following:

- Choose Highlight > Highlight BW (Option-Shift-H) from the Viewer's option menu.

To make high-contrast black and white the default highlight:

- 1 Open the Preferences > User > Color window.
- 2 Check "Mattes display high contrast black and white."
- 3 Click Save.

The next time you click the Highlight control in the Qualifier palette, the high-contrast highlight will display.

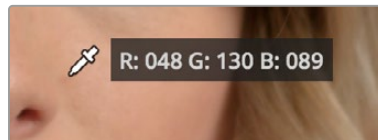
Using Highlight to Solo Nodes

There's one other use of the Highlight control, and that's to solo individual nodes in the Viewer and video out, rather than outputting the sum of the entire node tree. For example, if you're trying to adjust a stack of nodes attached to a Parallel or Layer Mixer node, it's difficult to see what you're doing while adjusting one of the stack of nodes, because only the final result is being output.

However, if you turn on Highlight while you're selecting a node with no key, then only the currently selected node is output, making it easy to view just one node as you adjust it. Just make sure you turn Highlight off before moving on, or you may confuse yourself.

Showing Picker RGB Values

While you're dragging the pointer over the Viewer and looking for a feature to sample, you can enable a tooltip that shows you the RGB Value of the pixel under the pointer by right-clicking the Viewer and choosing Show Picker RGB Value to toggle this feature on and off. When you turn this feature on, the Show RGB Picker Values In submenu in the Viewer's option menu has options for displaying either 8- or 10-bit tristimulus values.



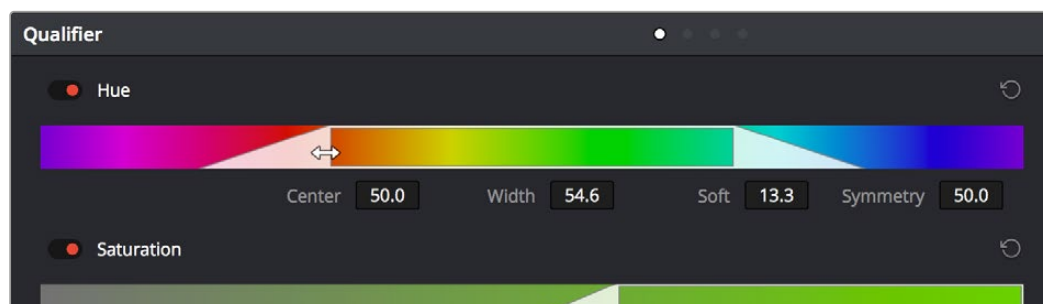
The Color Picker tooltip that appears when you turn on show RGB Picker Values

Qualifier Parameters

Each of the Qualifier modes, except for the 3D qualifier, has parameters that correspond to the relevant color components for that mode, with which you can identify a range and softness, similar to using the sample tools. These parameters can be edited precisely with virtual sliders using the pointer, or using the knobs of a supported grading control panel. However, mouse and tablet users can take advantage of each qualifier's draggable white overlays, for more direct adjustment.

- Drag the left and right edges of any qualifier control overlay to adjust the Low and High values (or the Width value of the Hue control).
- Drag the center of any qualifier control to change the center or to simultaneously change the Low and High parameters together.
- Option-drag the left and right edges of any qualifier overlay to adjust softness.

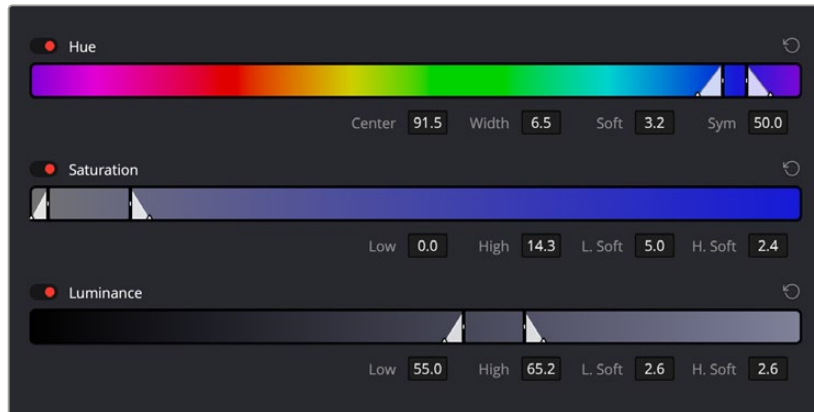
These draggable controls work for the HSL, RGB, and LUMA qualifier controls (the 3D qualifier works differently).



Draggable qualifier controls

HSL Qualification Controls

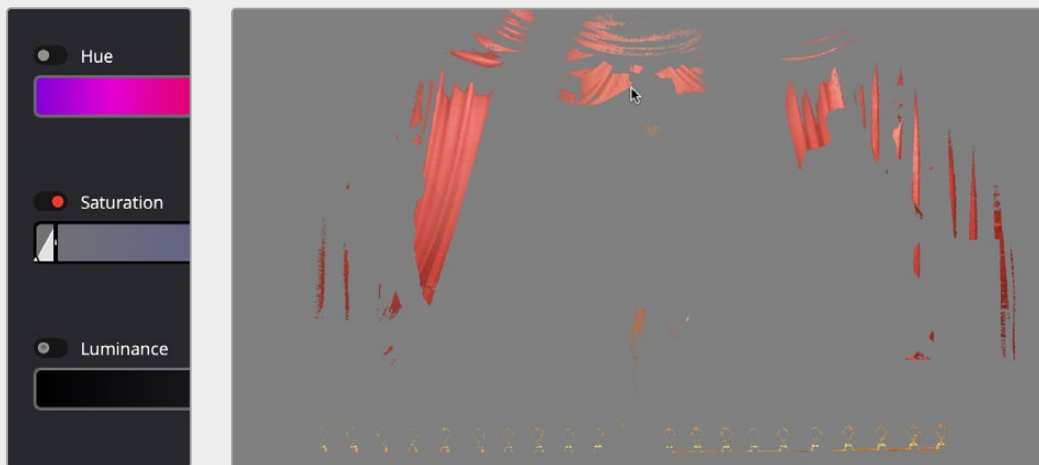
The HSL Qualifier is by far the most versatile mode of the Qualifier palette. It's also the qualifier that's easiest to understand, and the easiest to adjust by hand if you don't feel the need to sample the image. By selecting range and softness for each of three different color components, hue, saturation, and luma, you can isolate a wide variety of colorful subjects, or ranges of image lightness and saturation.



The HSL Qualifier controls in close up

One of the HSL Qualifier's key strengths, and one of its most underutilized features, is that you can enable or disable each of the three HSL components using the Hue, Sat, and Lum buttons, using toggle controls found at the left of each qualifier range control's name. This lets you ignore specific color components, while focusing on others that are more important.

For example, if you're trying to isolate the saturated parts of an image, regardless of the hue or brightness, you can turn off the Hue and Lum qualifiers so that only Sat is used to sample the image.



Qualification with saturation only to isolate the most saturated portion of the picture

The HSL parameters include:

- **Hue Center:** Defines the center of the range of Hue being isolated.
- **Hue Width:** Defines the width of the range of hue being isolated. Widening or narrowing this parameter widens and narrows the range of hue about the current Hue Center.
- **Hue Soft:** Widens or narrows the falloff at both sides of the currently selected range of hue.
- **Hue Symmetry:** Lets you selectively alter the angle of Hue Soft falloff at either the right or left side. Lowering this parameter makes the softness to the right of the hue range grow steeper. Raising this parameter makes the softness to the left of the hue range grow steeper.
- **Sat Low/High:** Two parameters let you define the upper and lower range of saturation being isolated.

- **Sat Low Soft/High Soft:** Two parameters let you define the upper and lower softness falling off outside the current range of saturation.
- **Lum Low/High:** Two parameters let you define the upper and lower range of luma being isolated.
- **Lum Low Soft/High Soft:** Two parameters let you define the upper and lower softness falling off outside the current range of luma.

RGB Qualification Controls

The RGB Qualifier mode lets you isolate range and softness in the Red, Green, and Blue channels of an image. While this is not necessarily an intuitive way of manually creating a qualification from scratch, it can provide a different way of isolating potentially trickier ranges of the RGB color space.

If you start by sampling the image, then you can widen and narrow each color channel's isolated range of values. In some cases, you may find it's faster to isolate a specific range of continuous color by adjusting these controls.



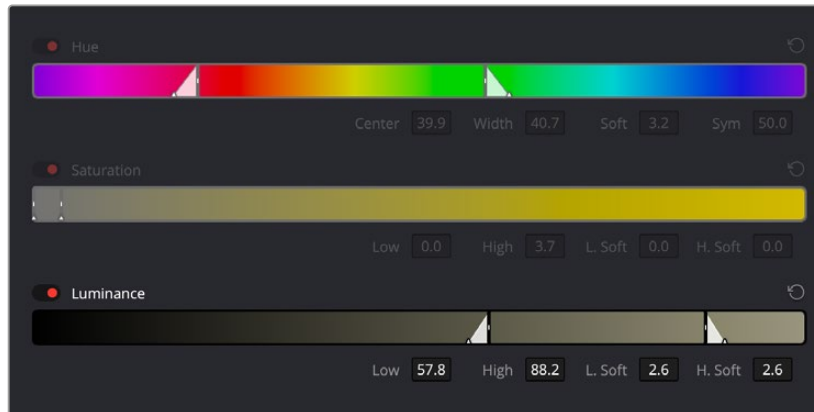
RGB qualification controls

The RGB parameters include:

- **Red Low/High:** Two parameters let you define the upper and lower range of the Red channel being isolated.
- **Red Low Soft/High Soft:** Two parameters let you define the upper and lower softness falling off outside the current range of red.
- **Blue Low/High:** Two parameters let you define the upper and lower range of the Blue channel being isolated.
- **Blue Low Soft/High Soft:** Two parameters let you define the upper and lower softness falling off outside the current range of blue.
- **Green Low/High:** Two parameters let you define the upper and lower range of the Green channel being isolated.
- **Green Low Soft/High Soft:** Two parameters let you define the upper and lower softness falling off outside the current range of green.

Luma Qualification Controls

The Luma Qualifier mode simply pulls a key from the Luma channel all by itself. It's identical to the HSL qualifier with H and S turned off. This single-component keyer is more useful than you might think; it's a common technique to isolate a range of highlights, midtones, or shadows throughout the image to alter the color temperature selectively.



The Luma Keyer mode automatically disables the Hue and Saturation controls

Also, keep in mind that the luma component is the sharpest keyable component when you're working with heavily compressed video. Keys that are noisy or chunky with HSL qualification may be sharper if you just use the Luma Qualifier, although you won't be able to be chromatically specific.

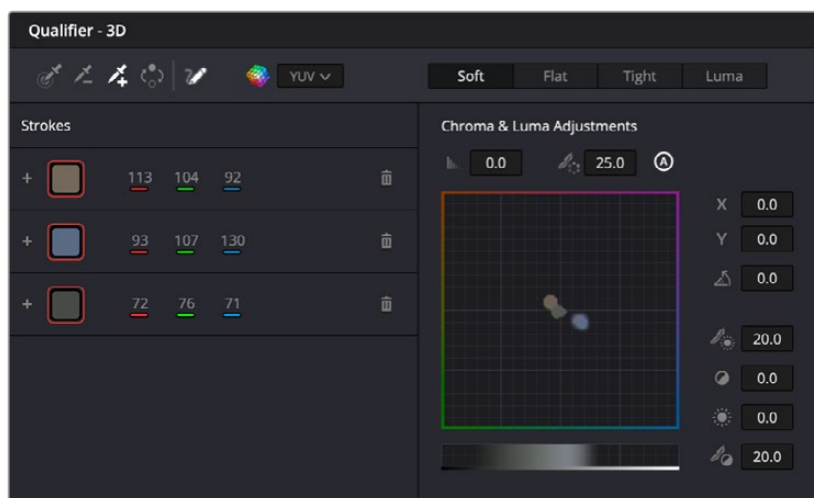
The Luma Qualifier has four parameters:

Lum Low/High: Two parameters let you define the upper and lower range of the luma component being isolated.

Lum Low Soft/High Soft: Two parameters let you define the upper and lower softness falling off outside the current range of luma.

3D Qualifier Controls

As discussed earlier in this chapter, the 3D Qualifier works much differently than the other qualifiers in DaVinci Resolve. In this mode, drawing lines over parts of the image you want to isolate adds samples to the Sample list, some of which add to the key (when you drag with the pointer) and some of which subtract from the key (when you Option-drag with the pointer). While this functionality is very simple to use, there are other controls that are available.



The 3D Qualifier mode

The 3D Stroke Pickers:



Picker: Chooses the initial color to qualify. Longer strokes will tend to give you a better key.



Picker -: Removes a color region from the qualifier; also available by holding down the Option key.



Picker +: Choses an additional color region to add to the initial qualifier.



Invert: Inverts the current key; qualified areas are then unqualified and vice versa.



Show Paths: A checkbox that lets you turn the visibility of the lines you're drawing to sample the image on and off. Turning lines off does not affect the key in any way.



Auto-B/W Highlight: With this checkbox enabled, the viewer automatically switches to Highlight B/W mode while drawing a line, to better show you the resulting key in real time. When the button is released the viewer will switch back to its original viewing mode.



Colorspace: A drop-down menu that lets you choose whether you're sampling colors in YUV colorspace, or HSL, HSP, or LAB colorspaces. YUV is the default, but if you find you're not getting satisfactory results, you can reset the Qualifier palette and switch to different ones to see if that works any better.



Strokes list: A list shows each line you drew as a sampled entry. A color patch at the left shows you the average color you sampled; clicking the color patch lets you temporarily turn any color sample off to evaluate its contribution to the overall key. A plus or minus icon shows you whether that patch is adding to or subtracting from the key. The average red, green, and blue color values are shown next, and a trash can icon at the right lets you delete any sample from the list.

The 3D Qualifier Modes:



Soft: Provides a softer edge to the key that is more forgiving of Chroma and Luma adjustments. The more gentle drop-off is suitable for situations like subtle light changes across a face. Use the Shadow / Midtone / Highlight Matte Finesse Controls in conjunction with this mode to fine tune the result.



Flat: The default mode. Each color selected is 100% keyed, and adjustments are made for small color variations. This is the ideal mode for chroma keying a green or blue screen. Additionally, turning up the Pre-Filter setting in the Matte Finesse controls can make for a smoother, flatter key.



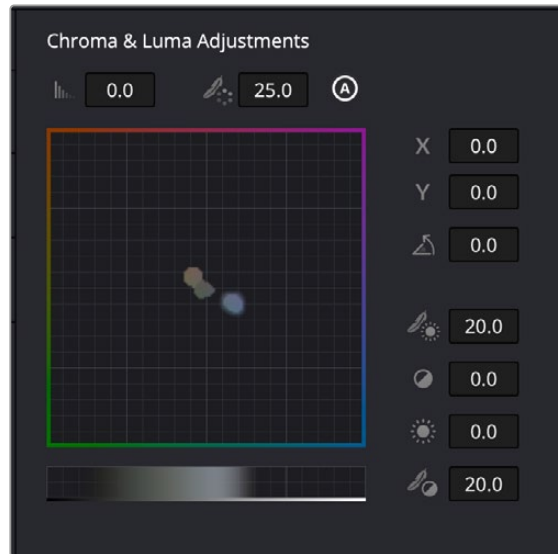
Tight: Only keys the exact color picked, and does not apply any softening based on color ranges. You chose that exact color, and only that exact color. Single pixel sharpness levels are expected. This can be used for difficult keying jobs, requiring adjusting the filter and softness manually in the Matte Finesse Controls.



Luma: Functions similar to the Tight setting but ignores all chroma data. This mode is used for black and white footage.



Despill: If you're using the 3D qualifier to pull a blue or green screen key to create transparency, this slider lets you adjust an automatic color correction that eliminates blue or green spill from the image, while retaining the image's original color.



The 3D Qualifier Chroma and Luma Adjustments showing the colors selected in the Strokes List.

The Chroma and Luma adjustment section is dominated by the Color Space box and the Brightness Range. The Color Space box is laid out roughly in the same manner as the Color Wheels, with primary colors laid out around its edge. The exact colors and layout is determined by the Color Space that you choose in the 3D Qualifier. The Brightness Range is represented as a bar underneath the Color Space, with left being black and right being white. Inside the Color Space and Brightness Range lie the individual colors that you chose with the pickers in the Strokes List. The tools around the periphery adjust the colors within the box in various ways, allowing you to visualize changes to the key.

The 3D Qualifier Chroma & Luma Adjustments:



Chroma Tolerance: Click and drag left and right to expand or contract the range of colors selected by the key.



Chroma Softness: Click and drag left and right to change the sensitivity to similar colors selected by the key. This determines whether the key is a hard cutoff or a soft selection of similar colors.



Adaptive Chroma Softness: Activating this icon allows the keyer to operate consistently in both highly saturated and desaturated regions of the image at the same time. The majority of the time you will want to leave this on. The exception is If you are having difficulty in manually adjusting chroma softness in Soft or Flat mode, then turning Adaptive Chroma Softness off will give you more range to work with. Adaptive Chroma Softness is automatically disabled in Tight and Luma modes, as it is contrary to the functionality of these modes.



X: Moves your selected color region within the color space, changing its chroma based on its horizontal position.



Y: Moves your selected color region within the color space, changing its chroma based on its vertical position.



Angle: Moves your selected color region within the color space, changing its chroma based on its rotation around the central origin point.



Low: Expands or contracts the dark areas of the key in the brightness range.



High: Expands or contracts the bright areas of the key in the brightness range.



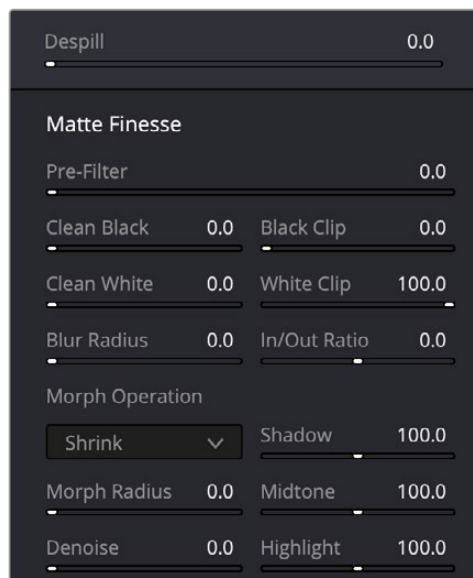
Low Soft: Controls how well defined the low end of the brightness range is. This determines whether the key is a hard cutoff or a soft selection of similar brightness levels.



High Soft: Controls how well defined the high end of the brightness range is. This determines whether the key is a hard cutoff or a soft selection of similar brightness levels.

Matte Finesse Controls

After you've used the sample controls and qualifier parameters to create the most useful key for your purposes, you may find that it has problems that cannot be overcome easily by the available adjustments. Issues such as chattery edges, holes, or noisy bits can sometimes be easily fixed using the Matte Finesse controls. These controls filter the output of the Qualifier controls and are adjustments that are made to the matte itself, so they may work well in some instances, and poorly in others, depending on what you're trying to do.



The Matte Finesse controls

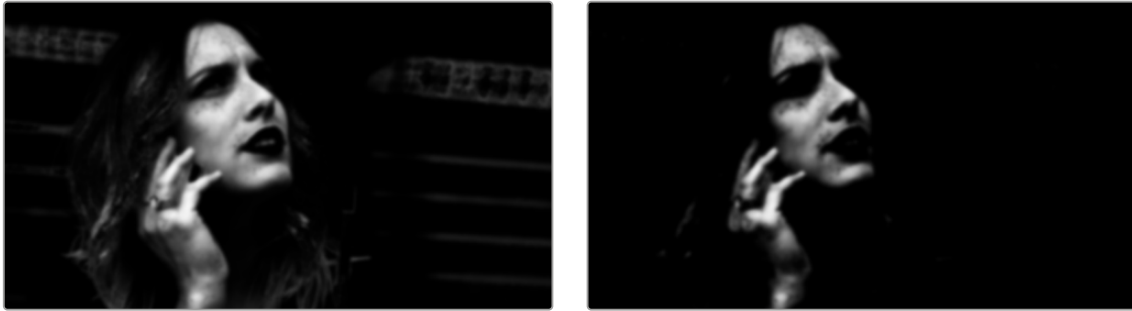
Pre-Filter

This slider attempts to clean up the image before colors are sampled. This adjustment can be useful when you have footage containing MPEG blocking artifacts.

Clean Black

Clean Black is a specialized operation that eliminates noise (seen as white speckling when viewing a high-contrast highlight) in the black area of a key that omits the regions of the image you don't want to isolate, and shrinks the key by making the dark parts of a key darker the higher you raise this

parameter, pushing dark gray areas of the key toward black. The practical result is that raising Clean Black lets you “fill holes” in the background portion of a key and erode translucent edges.



Before/after raising Clean Black, viewing the high contrast matte

Black Clip

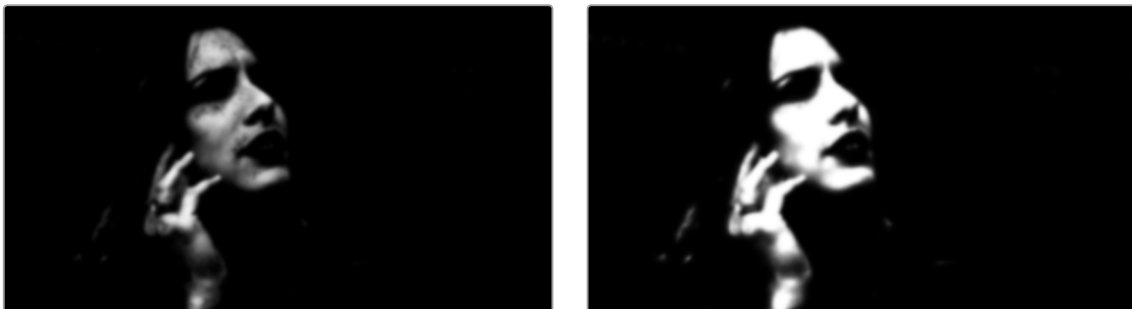
Raising Black Clip applies a “lift” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed towards black. The range is 0 to 100, with 0 being the default setting.



Before/after raising Black Clip

Clean White

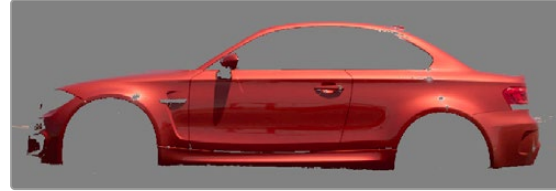
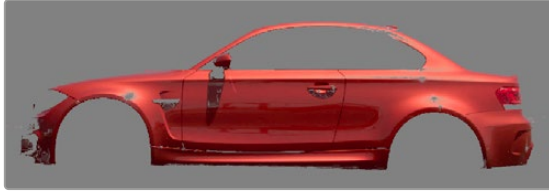
Clean White is another specialized operation that eliminates noise (seen as black speckling when viewing a high-contrast highlight) in the white portion of a key that include areas of the image you’re isolating, and expands the key by making light parts of a key lighter the higher you raise this parameter, pushing light gray areas of the key toward white. The practical result is that raising Clean White lets you “fill holes” in the foreground portion of a key and grow translucent edges.



Before/after raising Clean White, viewing the high contrast matte

White Clip

Lowering White Clip applies a “gain” adjustment such that translucent areas of the matte (gray areas when viewing a high-contrast highlight) are pushed towards white. The range is 0 to 100, with 100 being the default setting.

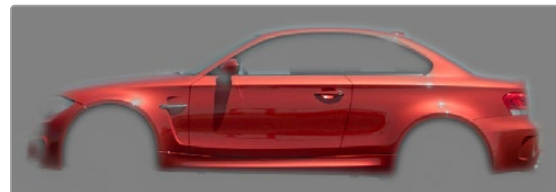
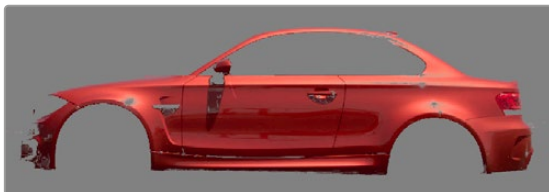


Before/after lowering White Clip

Blur Radius

In small amounts, blurring a key does well to take the edge off problem edges. However, blurring a key can also feather the edges of a key past the border of the subject you’re keying, with the result being a visible “halo” around your subject depending on the adjustment you’re making. The range is 0 to 2000, with 0 being the default. With such a large maximum blur radius, combined with the capabilities that the In/Out Ratio provides in customizing the direction of spread, you can turn some pretty precarious mattes into surprisingly smooth and useful results.

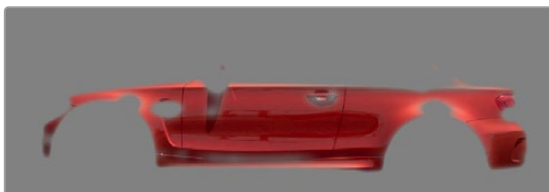
TIP: If haloing is a problem, you can try adjusting the In/Out Ratio parameter, or you may want to consider using the Qualifier Softness parameters to feather the edges of the key, instead.



Using the Blur parameters to soften the edges of a key

In/Out Ratio

Controls whether the “Blur Radius” is applied equally to the inside and outside of the edge of the matte (zero), applied only within the matte (negative values), or applied only outside the matte (positive values). Using In/Out Ratio can help eliminate fringing when using the Blur Radius parameter.



(Left) In/Out Ratio lowered so blurring ends at the edge of the matte and feathers inward,
(Right) In/Out Ratio raised so that blurring starts at the edge of the matte and feathers outward

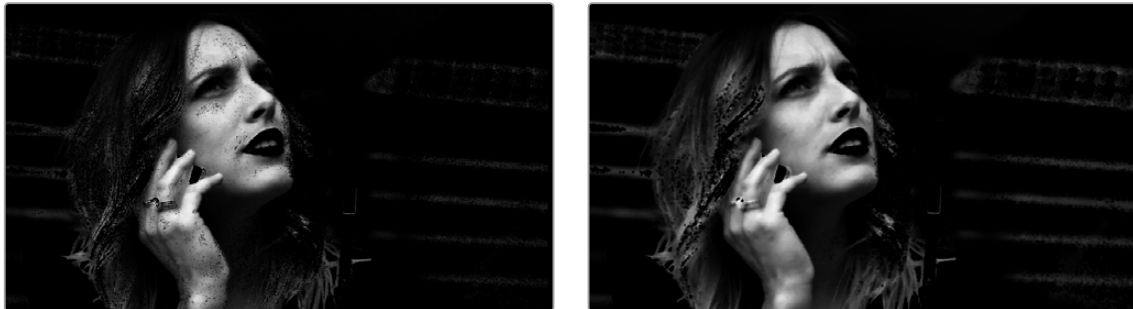
However, you can also use In/Out Ratio even in situations where no Blur Radius is applied at all. Raising In/Out Ratio will fill in small black holes in the matte, while lowering In/Out Ratio below 0 will eliminate speckling by pushing small white bits of the matte toward black.



(Left) In/Out Ratio controls lowered to expand holes in the matte, (Right) In/Out Ratio controls raised to fill in small black holes in the matte

Morph Operation

Lets you choose how you want to modify the alpha channel/key. You can choose Shrink or Grow to dilate or erode the edges of the matte with great accuracy. Or, you can choose Opening or Closing to plug or expand holes to clean up a ragged matte.



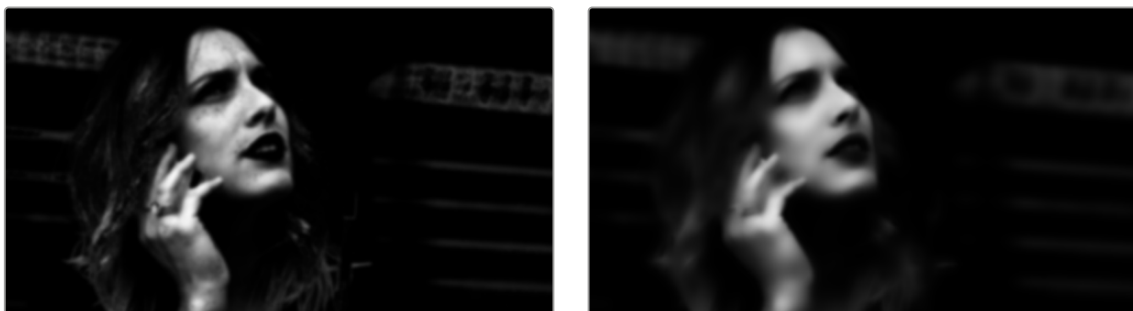
(Left) A matte with small holes in the face, (Right) Mode set to Shrink, Morph Radius set to 3; used to close holes in a matte

Morph Radius

Combined with Morph Operation, this slider adjusts how much to shrink, grow, open, or close the edge's key.

Denoise

Denoise provides a distinct way to post-process extracted keys to selectively reduce the noise in a key, getting rid of stray areas of qualification and softly filling holes in a matte.



Before/after raising Denoise, viewing the high contrast matte

Shadow

Adjusts key strength based on the darker parts of the original image.

Midtone

Adjusts key strength based on the midtones of the original image.

Highlight

Adjusts key strength based on the brighter parts of the original image.

Post-Filter

Performs a final clean-up of the key, using the original image for reference, useful for bringing back some fine detail in sharp edges or hair.

The Many Ways to Invert a Key

Sometimes, you'll need to isolate a subject or range of values in an image, and then make a change to everything else. One way to do this is to add an Outside node after the node in which you've created the key. This lets you make one set of adjustments to what you've keyed, and another set of adjustments to everything outside the key.

If you just need to make a simple adjustment, you can invert the key from within the Qualifier palette. However, there are three other ways to invert a key.

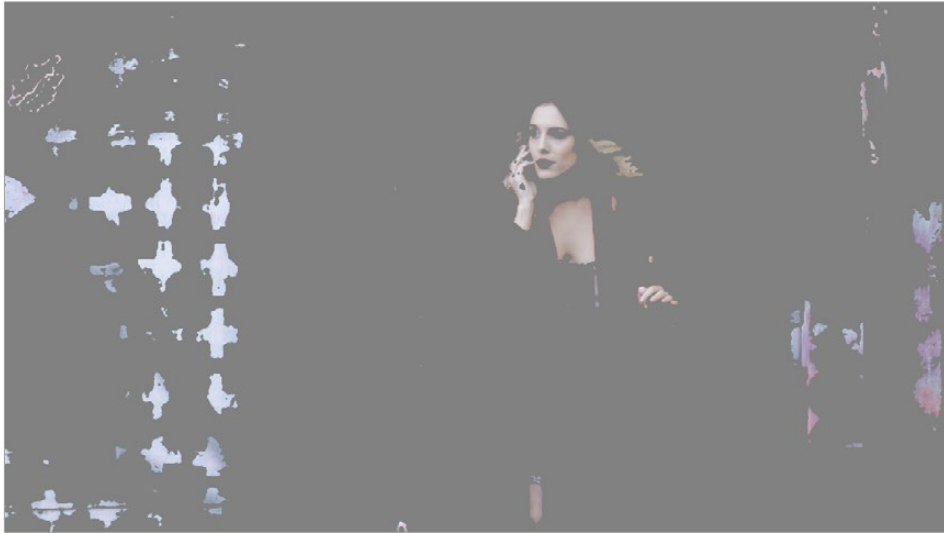
The Qualifier Invert button: Clicking this control simply inverts the key being generated by all of the controls within the Qualifier palette.

Key Palette Qualifier Invert: The Qualifier Invert control within the Key palette provides another control with which you can invert the key that's generated by the Qualifier palette.

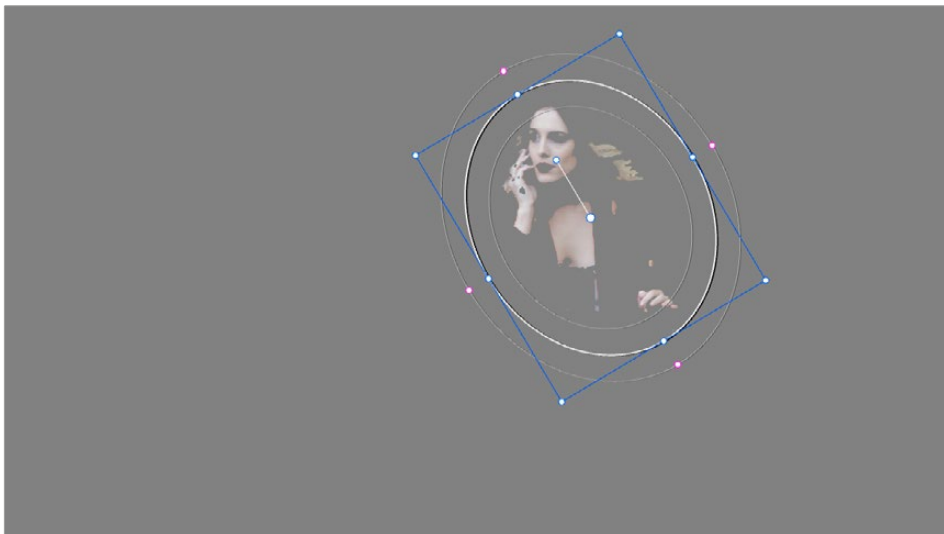
Key Palette Output Invert: The Output Invert control within the Key palette is notable because it inverts the sum of all keys applied within that node. For example, if you're using a Qualifier in conjunction with a Window, the two invert controls mentioned previously will only invert the Qualifier key; the Window key remains unaffected. However, using the Output Invert control inverts the overall combination of the Qualifier key and the Window, all together.

Combining Qualifiers and Windows

This is covered in more detail in a subsequent section, but if you create a key using one of the Qualifier modes, and you then add a Window, the final key that's output by that node is limited to the intersection of the Qualifier key and the Window. This makes it easy to use a Window to “garbage matte” out bits of a key that you don't want, that can't be eliminated by further adjustment of the Qualifier controls.



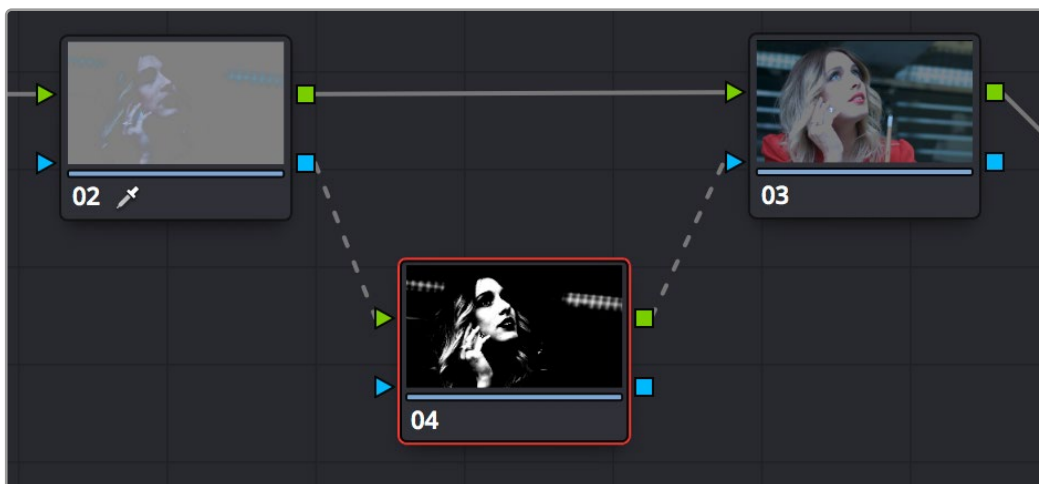
The woman's skin tone is too close to other elements in the scene.



A Circular Power Window is used to further isolate the woman being isolated.

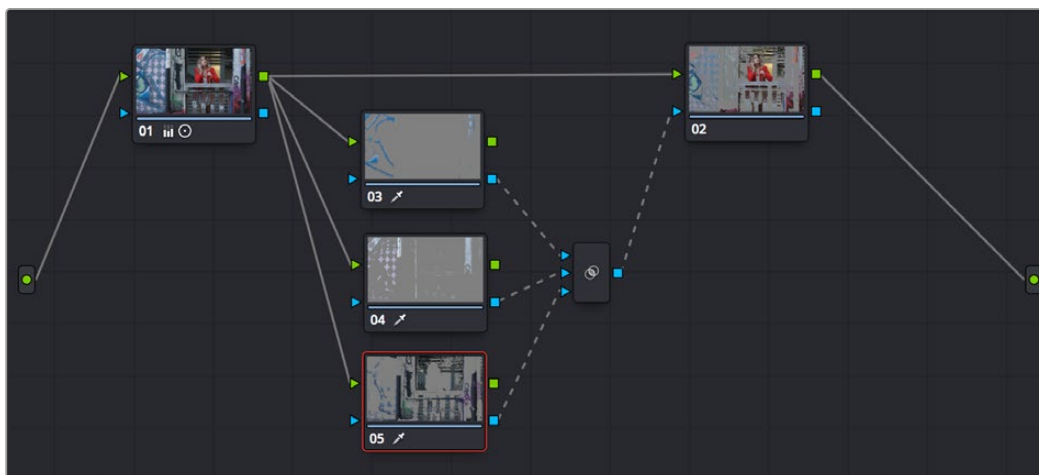
Manipulating Keys Using Additional Nodes

If you need to make more adjustments to a qualified key than the Matte Finesse controls will allow, you can use the Node Editor to feed the key output of one node to the RGB input of another, at which point you can use all of the second node's color adjustment controls to manipulate the grayscale image that constitutes that key, to improve it.



Connecting a key output to an RGB input, and then connecting the RGB output back to the next node's KEY input again

You can also use the Key Mixer node to combine multiple keys in several different ways, adding keys together, or subtracting them from one another, in order to create exactly the key you need.



Adding multiple keys together using a Key Mixer

For more information about these techniques, see *Chapter 146, "Combining Keys and Using Mattes."*